Numerical Analysis: Final Exam

(**50 marks**, only the 5 best questions count)

Solutions

Urbain Vaes

12 December 2022

You are allowed to use a calculator, but not *Julia* or *Python*.

Question 1 (Floating point arithmetic, **10 marks**)**.** True or false? +1/0/-1

1. Let $(\bullet)_3$ denote base 3 representation. It holds that
$$(222, 222)_3 + (1)_3 = (1, 000, 000)_3.$$

2. Let $(\bullet)_2$ denote base 2 representation. It holds that
$$3 \times (0.0101)_2 = (0.1111)_2.$$

3. The following equality holds
$$(0.\overline{011})_2 = \frac{3}{4}.$$

4. The number $x = (d_1 d_2 d_3)_3$ for $d_1, d_2, d_3 \in \{0, 1, 2\}$ is a multiple of 3 if and only if $d_3 = 0$.

5. In Julia, `Float64(0.375) == Float32(0.375)` evaluates to `true`.

6. The value of the machine epsilon is the same for the single precision ($\mathbf{F}_{32}$) and the double precision ($\mathbf{F}_{64}$) formats.

7. The spacing (in absolute value) between successive double-precision (`Float64`) floating point numbers is equal to the machine epsilon.

8. All the natural numbers can be represented exactly in the double precision floating point format $\mathbf{F}_{64}$.

9. Machine addition in the $\mathbf{F}_{64}$ format is associative but not commutative.

10. In Julia `exp(eps()) == 1 + eps()` evaluates to `true`. (Remember that, by default, rounding is to the nearest representable number).

11. In Julia `sqrt(1 + eps()) == 1 + eps()` evaluates to `true`.

12. Let x and y be two numbers in $\mathbf{F}_{64}$. The result of the machine multiplication $x \mathbin{\widehat{*}} y$ is sometimes exact and sometimes not, depending on the values of x and y.

13. In Julia, let `f(x) = (x == x/100.0) ? x : f(x/100.0)` [1]. Then `f(3.0)` returns `0.0`.

Solution. Below, $\varepsilon = 2^{-52}$ denotes the machine epsilon for $\mathbf{F}_{64}$.

1. **True**: $(222, 222)_3 = 3^6 - 1$, so adding 1 gives $3^6 = (1, 000, 000)_3$.
2. **True**: $3 \times (0.0101)_2 = 3 \times \frac{5}{16} = \frac{15}{16} = (0.1111)_2$.
3. **False**: if $x = (0.\overline{011})_2$, then $2^3 x = (11)_2 + x = 3 + x$, so $x = \frac{3}{7}$.
4. **True**: $x = 9d_1 + 3d_2 + d_3$, so x is a multiple of 3 if and only if $d_3 \in \{0, 1, 2\}$ is, i.e. if and only if $d_3 = 0$.
5. **True**: $0.375 = \frac{3}{8} = (0.011)_2$ is exactly representable in both formats.
6. **False**: the machine epsilon is 2^{-23} for $\mathbf{F}_{32}$ and 2^{-52} for $\mathbf{F}_{64}$.
7. **False**: the spacing between successive numbers in $[2^e, 2^{e+1})$ is $2^e\varepsilon$, so it depends on the magnitude of the numbers.
8. **False**: for example, $2^{53} + 1$ is not representable in $\mathbf{F}_{64}$.
9. **False**: it is the opposite: machine addition is commutative but not associative. For example, $(1 \mathbin{\widehat{+}} \varepsilon/2) \mathbin{\widehat{+}} \varepsilon/2 = 1$, whereas $1 \mathbin{\widehat{+}} (\varepsilon/2 \mathbin{\widehat{+}} \varepsilon/2) = 1 + \varepsilon$.
10. **True**: $\exp(\varepsilon) = 1 + \varepsilon + \frac{\varepsilon^2}{2} + \cdots$ is much closer to $1 + \varepsilon$ than to the neighbouring floating point numbers 1 and $1 + 2\varepsilon$.
11. **False**: $\sqrt{1 + \varepsilon} = 1 + \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8} + \cdots$ is slightly smaller than the midpoint $1 + \frac{\varepsilon}{2}$ between 1 and $1 + \varepsilon$, so it is rounded to 1.
12. **True**: for example, $2 \mathbin{\widehat{*}} 3 = 6$ is exact, whereas $0.1 \mathbin{\widehat{*}} 0.1$ is not, because the exact product of two numbers with 53 significant bits may require up to 106 significant bits.
13. **True**: the condition `x == x/100.0` holds only for $x = 0$ (among finite numbers). Dividing 3 repeatedly by 100, we obtain a number that underflows to 0 after finitely many steps, and then the function returns `0.0`.

△

Question 2 (Interpolation, **10 marks**). Let $u\colon [-1, 1] \to \mathbf{R}$ be given by

$$u(x) = x^3.$$

[1] In Python, let `f = lambda x: x if x == x/100.0 else f(x/100.0)`

Let $p\colon [-1,1] \to \mathbf{R}$ denote the interpolating polynomial of u at nodes $x_0 < x_1 < x_2$, all contained in the interval $[-1,1]$.

1. (**2 marks**) Let $e(x) := u(x) - p(x)$. Prove, without assuming any result shown in class, that the interpolation error satisfies

$$\forall x \in [0,1], \qquad e(x) = (x - x_0)(x - x_1)(x - x_2).$$

2. (**2 marks**) Using a method of your choice, calculate the interpolating polynomial p in the particular case where

$$x_0 = -1, \qquad x_1 = 0, \qquad x_2 = 1. \tag{1}$$

3. (**2 marks**) We denote the maximum absolute value of the error by

$$E := \max_{x \in [-1,1]} \big|e(x)\big|. \tag{2}$$

   Calculate the value of E in the particular case (1).

4. (**2 marks**) We denote by $T_3\colon [-1,1] \to \mathbf{R}$ the Chebyshev polynomial given by

$$T_3(x) := \cos\big(3 \arccos(x)\big).$$

   Show that

$$T_3(x) = 4x^3 - 3x$$

   and calculate the roots z_0, z_1, z_2 of T_3.

   Hint: Note that $\cos(3\theta) = \Re\left(\mathrm{e}^{\mathrm{i}3\theta}\right) = \Re\big(\big(\mathrm{e}^{\mathrm{i}\theta}\big)^3\big)$, where $\mathrm{e}^{i\theta} = \cos(\theta) + \mathrm{i}\sin(\theta)$.

5. (**2 marks**) Find the expression of the error $e(x)$ and the maximum absolute error E given in (2) in the case where the interpolation nodes x_0, x_1, x_2 are given by z_0, z_1, z_2.

6. *(**Bonus +2**) Show that the maximum absolute error (2), viewed as a function of the interpolation nodes x_0, x_1, x_2, is minimized when $x_i = z_i$ for $i \in \{0,1,2\}$.

   Hint: Reason by contradiction and notice that

$$\big|T_3(y)\big| = 1 \qquad \text{for } y \in \left\{-1, -\frac{1}{2}, \frac{1}{2}, 1\right\}.$$

Solution. **1.** Since $p \in \mathcal{P}_2$, the error $e = u - p$ is a polynomial of degree 3 with leading coefficient 1. In addition, it vanishes at x_0, x_1 and x_2 by definition of p. A polynomial of degree 3 with three distinct roots x_0, x_1, x_2 is of the form $c(x-x_0)(x-x_1)(x-x_2)$, and $c = 1$ by comparing the leading coefficients. Therefore $e(x) = (x-x_0)(x-x_1)(x-x_2)$ for all $x \in \mathbf{R}$, and in particular on $[0,1]$.

2. By the previous item, $p(x) = u(x) - (x+1)x(x-1) = x^3 - (x^3 - x) = x$.

3. In this case $e(x) = x^3 - x$, which is an odd function. Its derivative $3x^2 - 1$ vanishes at $x = \pm\frac{1}{\sqrt{3}}$, and $e(\pm 1) = 0$, so

$$E = \left| e\left(\frac{1}{\sqrt{3}}\right)\right| = \frac{1}{\sqrt{3}}\left(1 - \frac{1}{3}\right) = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.385.$$

4. Let $\theta = \arccos(x)$, so that $x = \cos(\theta)$ and $\sin(\theta)^2 = 1 - x^2$. By the binomial formula,

$$\cos(3\theta) = \Re\big((\cos\theta + \mathrm{i}\sin\theta)^3\big) = \cos(\theta)^3 - 3\cos(\theta)\sin(\theta)^2 = x^3 - 3x(1-x^2) = 4x^3 - 3x.$$

The roots satisfy $3\arccos(x) \in \left\{\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}\right\}$, so $z_0 = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$, $z_1 = \cos\left(\frac{\pi}{2}\right) = 0$ and $z_2 = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$.

5. By the first item, e is the monic polynomial of degree 3 with roots z_0, z_1, z_2, that is

$$e(x) = x\left(x^2 - \frac{3}{4}\right) = \frac{1}{4}T_3(x).$$

Since $|T_3(x)| = |\cos\big(3\arccos(x)\big)| \leqslant 1$, with equality at $x = 1$, we deduce that $E = \frac{1}{4}$.

6. For given nodes, let $\omega(x) = (x-x_0)(x-x_1)(x-x_2)$, so that $E = \max_{[-1,1]}|\omega|$ by the first item. Suppose for contradiction that $\max_{[-1,1]}|\omega| < \frac{1}{4}$. The function $d = \frac{1}{4}T_3 - \omega$ is a polynomial of degree at most 2, because both $\frac{1}{4}T_3$ and ω are monic of degree 3. Since $\frac{1}{4}T_3$ takes the values $-\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, \frac{1}{4}$ at the points $y = -1, -\frac{1}{2}, \frac{1}{2}, 1$, the polynomial d takes nonzero values of alternating signs at these four points. By the intermediate value theorem, d has at least three roots, so $d = 0$ since it has degree at most 2. Then $\omega = \frac{1}{4}T_3$, which contradicts $\max_{[-1,1]}|\omega| < \frac{1}{4}$. Therefore $E \geqslant \frac{1}{4}$ for any choice of nodes, with equality when $x_i = z_i$.

△

Question 3 (Numerical integration, **10 marks**)**.** Let $u\colon [0,1] \to \mathbf{R}$ be a function we wish to integrate and

$$I := \int_0^1 u(x)\,\mathrm{d}x.$$

1. (**3 marks**) Consider the following integration rule:

$$I \approx w_1 u(0) + w_2 u(1). \tag{3}$$

Find the weights $w_1, w_2 \in \mathbf{R}$ so that this integration rule has the highest possible degree of precision. What is the degree of precision of the rule constructed?

2. (**3 marks**) Let $x_i = i/n$ for $i = 0, \ldots, n$. The composite trapezoidal rule is given by

$$I \approx \frac{1}{2n}\big(u(x_0) + 2u(x_1) + 2u(x_2) + \cdots + 2u(x_{n-2}) + 2u(x_{n-1}) + u(x_n)\big) =: \widehat{I}_n. \tag{4}$$

Explain how this rule can be obtained by applying a generalization of the integration rule (3) in each interval $[x_i, x_{i+1}]$.

3. (**3 marks**) Assume that $u \in C^2\big([0,1]\big)$. Show that, for all $n \in \mathbf{N}_{>0}$,

$$\big|I - \widehat{I}_n\big| \leqslant \frac{C_2}{12n^2}, \qquad C_2 := \sup_{\xi\in[0,1]} \big|u''(\xi)\big|. \tag{5}$$

You may use Theorem 1 at the end of this document for the interpolation error.

4. (**1 mark**) In this part of the question, we assume that u is a quadratic polynomial. It is possible to show that, in this case,

$$I - \widehat{I}_n = -\frac{u''(0)}{12n^2}.$$

Explain how, given two approximations $\widehat{I}_n$ and $\widehat{I}_{2n}$ obtained with (4), a better approximation of the integral I can be obtained by a linear combination of the form

$$\alpha_1 \widehat{I}_n + \alpha_2 \widehat{I}_{2n}.$$

5. *(**Bonus +2**) Instead of (3), consider a more general integration rule of the form

$$\int_0^1 u(x)\,\mathrm{d}x \approx w_1 u(x_1) + w_2 u(x_2). \tag{6}$$

Find the weights $w_1, w_2 \in \mathbf{R}$ and the nodes $x_1, x_2 \in [0,1]$ so that this integration rule has the highest possible degree of precision. What is the degree of precision obtained?

Solution. **1.** Exactness for $u(x) = 1$ and $u(x) = x$ gives $w_1 + w_2 = 1$ and $w_2 = \frac{1}{2}$, so $w_1 = w_2 = \frac{1}{2}$ (trapezoidal rule). For $u(x) = x^2$, the rule gives $\frac{1}{2} \neq \frac{1}{3}$, so the degree of precision is 1. It is not possible to do better, because two conditions already determine the weights.

2. By the change of variable $x = a + (b-a)t$, the rule (3) generalizes to an interval $[a, b]$ as $\int_a^b u(x)\,\mathrm{d}x \approx \frac{b-a}{2}\big(u(a) + u(b)\big)$. Applying this rule in each interval $[x_i, x_{i+1}]$, of length

$\frac{1}{n}$, and summing, we obtain

$$I = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} u(x)\,\mathrm{d}x \approx \sum_{i=0}^{n-1} \frac{1}{2n}\big(u(x_i) + u(x_{i+1})\big) = \widehat{I}_n,$$

since each interior node belongs to two intervals.

3. Let $h = \frac{1}{n}$. In each interval, the trapezoidal rule integrates exactly the linear interpolant $\widehat{u}$ of u at x_i and x_{i+1}. By Theorem 1,

$$\left| \int_{x_i}^{x_{i+1}} u(y) - \widehat{u}(y)\,\mathrm{d}y \right| \leqslant \frac{C_2}{2} \int_{x_i}^{x_{i+1}} (y - x_i)(x_{i+1} - y)\,\mathrm{d}y = \frac{C_2}{2} \times \frac{h^3}{6} = \frac{C_2 h^3}{12}.$$

Summing over the n intervals gives $\left|I - \widehat{I}_n\right| \leqslant n\frac{C_2 h^3}{12} = \frac{C_2}{12n^2}$.

4. We have $I - \widehat{I}_n = -\frac{c}{n^2}$ and $I - \widehat{I}_{2n} = -\frac{c}{4n^2}$, with $c = \frac{u''(0)}{12}$. Multiplying the second equation by 4 and subtracting the first one, we eliminate c and obtain $3I = 4\widehat{I}_{2n} - \widehat{I}_n$, that is

$$I = -\frac{1}{3}\widehat{I}_n + \frac{4}{3}\widehat{I}_{2n}.$$

This combination (Richardson extrapolation) is exact for quadratic polynomials, and for a general smooth function it gives a more accurate approximation than $\widehat{I}_{2n}$.

5. This is the Gauss–Legendre rule with two nodes, which has degree of precision 3, the highest possible with two nodes. On $[-1, 1]$, the nodes are $\pm\frac{1}{\sqrt{3}}$ and the weights are equal to 1; mapping to $[0, 1]$ by $x = \frac{1+t}{2}$, we obtain

$$x_{1,2} = \frac{1}{2} \mp \frac{1}{2\sqrt{3}}, \qquad w_1 = w_2 = \frac{1}{2}.$$

One can also find these values directly: by symmetry, take $w_1 = w_2 = \frac{1}{2}$ and $x_{1,2} = \frac{1}{2} \mp s$; the rule is then exact for 1 and for odd powers of $x - \frac{1}{2}$, and exactness for $(x - \frac{1}{2})^2$ requires $s^2 = \int_0^1 (x - \frac{1}{2})^2\,\mathrm{d}x = \frac{1}{12}$. The rule is not exact for $(x - \frac{1}{2})^4$, so the degree of precision is 3.

△

Question 4 (Iterative method for linear systems, **10 marks**)**.** Assume that $\mathsf{A} \in \mathbf{R}^{n\times n}$ is a *symmetric positive definite* matrix and that $\boldsymbol{b} \in \mathbf{R}^n$. We wish to solve the linear system

$$\mathsf{A}\boldsymbol{x} = \boldsymbol{b}. \tag{7}$$

To this end we consider an iterative method where each iteration is of the form

$$\mathsf{M}\boldsymbol{x}_{k+1} = \mathsf{N}\boldsymbol{x}_k + \boldsymbol{b}. \tag{8}$$

Here $\mathsf{A} = \mathsf{M} - \mathsf{N}$ is a splitting of A such that M is nonsingular, and $\boldsymbol{x}_k \in \mathbf{R}^n$ denotes the k-th iterate of the numerical scheme.

1. (**3 marks**) Let $\boldsymbol{e}_k := \boldsymbol{x}_k - \boldsymbol{x}_*$, where $\boldsymbol{x}_*$ is the exact solution to (7). Prove that

$$\forall k \in \mathbf{N}, \qquad \boldsymbol{e}_{k+1} = \mathsf{M}^{-1}\mathsf{N}\boldsymbol{e}_k.$$

2. (**2 marks**) We denote by $\|\bullet\|_{\mathsf{A}}$ the vector norm

$$\|\boldsymbol{x}\|_{\mathsf{A}} := \sqrt{\boldsymbol{x}^T\mathsf{A}\boldsymbol{x}}, \tag{9}$$

and we use the same notation for the induced matrix norm. Prove that

$$\forall k \in \mathbf{N}, \qquad \|\boldsymbol{e}_k\|_{\mathsf{A}} \leqslant L^k\|\boldsymbol{e}_0\|_{\mathsf{A}}, \qquad L := \|\mathsf{M}^{-1}\mathsf{N}\|_{\mathsf{A}}. \tag{10}$$

3. (**1 mark**) Is the condition $\|\mathsf{M}^{-1}\mathsf{N}\|_{\mathsf{A}} < 1$ sufficient to ensure convergence for all $\boldsymbol{x}_0$?

4. *(**3 marks**) Show that

$$\|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x}\|_{\mathsf{A}}^2 = \|\boldsymbol{x}\|_{\mathsf{A}}^2 - \boldsymbol{y}^T(\mathsf{M}^T + \mathsf{N})\boldsymbol{y}, \qquad \boldsymbol{y} := \mathsf{M}^{-1}\mathsf{A}\boldsymbol{x}. \tag{11}$$

Hint: Eliminate N from both sides of the equation by rewriting $\mathsf{N} = \mathsf{M} - \mathsf{A}$. Then substitute the expression of $\boldsymbol{y}$ and expand both sides. Remember that a scalar quantity transposed is equal to itself.

5. (**1 mark**) Show that, for the Gauss–Seidel method, i.e. when $\mathsf{M} = \mathsf{L} + \mathsf{D}$ contains just the lower triangular and diagonal parts of A, it holds that

$$\mathsf{M}^T + \mathsf{N} = \mathsf{D}. \tag{12}$$

6. (**Bonus +2**) Deduce from (11) and (12) that, for the Gauss–Seidel method,

$$\|\mathsf{M}^{-1}\mathsf{N}\|_{\mathsf{A}} < 1.$$

Solution. **1.** Since $\mathsf{A}\boldsymbol{x}_* = \boldsymbol{b}$, the exact solution satisfies $\mathsf{M}\boldsymbol{x}_* = \mathsf{N}\boldsymbol{x}_* + \boldsymbol{b}$. Subtracting this from (8), we obtain $\mathsf{M}\boldsymbol{e}_{k+1} = \mathsf{N}\boldsymbol{e}_k$, and so $\boldsymbol{e}_{k+1} = \mathsf{M}^{-1}\mathsf{N}\boldsymbol{e}_k$.

2. By definition of the induced matrix norm, $\|\boldsymbol{e}_{k+1}\|_{\mathsf{A}} = \|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{e}_k\|_{\mathsf{A}} \leqslant L\|\boldsymbol{e}_k\|_{\mathsf{A}}$, and the result follows by induction.

3. Yes: if $L < 1$, then $\|\boldsymbol{e}_k\|_{\mathsf{A}} \leqslant L^k\|\boldsymbol{e}_0\|_{\mathsf{A}} \to 0$ for all $\boldsymbol{x}_0$.

4. Since $\mathsf{N} = \mathsf{M} - \mathsf{A}$, we have $\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x} = \boldsymbol{x} - \mathsf{M}^{-1}\mathsf{A}\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{y}$. Therefore, using that A is

symmetric,

$$\|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x}\|_{\mathsf{A}}^2 = (\boldsymbol{x}-\boldsymbol{y})^T\mathsf{A}(\boldsymbol{x}-\boldsymbol{y}) = \|\boldsymbol{x}\|_{\mathsf{A}}^2 - 2\boldsymbol{y}^T\mathsf{A}\boldsymbol{x} + \boldsymbol{y}^T\mathsf{A}\boldsymbol{y}.$$

Since $\mathsf{A}\boldsymbol{x} = \mathsf{M}\boldsymbol{y}$, and since $\boldsymbol{y}^T\mathsf{M}\boldsymbol{y} = (\boldsymbol{y}^T\mathsf{M}\boldsymbol{y})^T = \boldsymbol{y}^T\mathsf{M}^T\boldsymbol{y}$, it holds that $2\boldsymbol{y}^T\mathsf{A}\boldsymbol{x} = 2\boldsymbol{y}^T\mathsf{M}\boldsymbol{y} = \boldsymbol{y}^T(\mathsf{M}+\mathsf{M}^T)\boldsymbol{y}$. Consequently,

$$\|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x}\|_{\mathsf{A}}^2 = \|\boldsymbol{x}\|_{\mathsf{A}}^2 - \boldsymbol{y}^T(\mathsf{M}+\mathsf{M}^T-\mathsf{A})\boldsymbol{y} = \|\boldsymbol{x}\|_{\mathsf{A}}^2 - \boldsymbol{y}^T(\mathsf{M}^T+\mathsf{N})\boldsymbol{y}.$$

5. Since A is symmetric, it can be decomposed as $\mathsf{A} = \mathsf{L}+\mathsf{D}+\mathsf{L}^T$, where L is its strictly lower triangular part. For the Gauss–Seidel method, $\mathsf{M} = \mathsf{L}+\mathsf{D}$ and $\mathsf{N} = \mathsf{M}-\mathsf{A} = -\mathsf{L}^T$. Therefore $\mathsf{M}^T+\mathsf{N} = \mathsf{L}^T+\mathsf{D}-\mathsf{L}^T = \mathsf{D}$.

6. The diagonal entries of A are positive, because $a_{ii} = \boldsymbol{e}_i^T\mathsf{A}\boldsymbol{e}_i > 0$, so D is symmetric positive definite. For $\boldsymbol{x} \neq \boldsymbol{0}$, the vector $\boldsymbol{y} = \mathsf{M}^{-1}\mathsf{A}\boldsymbol{x}$ is nonzero, so $\boldsymbol{y}^T\mathsf{D}\boldsymbol{y} > 0$ and, by (11) and (12),

$$\|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x}\|_{\mathsf{A}} < \|\boldsymbol{x}\|_{\mathsf{A}}.$$

The function $\boldsymbol{x} \mapsto \|\mathsf{M}^{-1}\mathsf{N}\boldsymbol{x}\|_{\mathsf{A}}$ is continuous on the compact set $\{\boldsymbol{x} : \|\boldsymbol{x}\|_{\mathsf{A}} = 1\}$, so it attains its maximum there, which is therefore strictly less than 1. Hence $\|\mathsf{M}^{-1}\mathsf{N}\|_{\mathsf{A}} < 1$. △

Question 5 (Nonlinear equations, **10 marks**)**.** We consider the following iterative method for calculating $\sqrt[3]{2}$:

$$x_{k+1} = F(x_k) := \omega x_k + (1-\omega)\frac{2}{x_k^2}, \tag{13}$$

with $\omega \in (0,1)$ a fixed parameter.

1. (**1 mark**) Show that $x_* := \sqrt[3]{2}$ is a fixed point of the iteration (13).

2. (**2 marks**) Write down in pseudocode a computer program based on the iteration (13) for calculating $\sqrt[3]{2}$. Use an appropriate stopping criterion that does not require to know the value of $\sqrt[3]{2}$.

3. (**2 marks**) Prove that if $\omega \in \left(\frac{1}{3}, 1\right)$, then x_* is locally exponentially stable. You may take for granted Theorem 2 at the end of this document.

4. (**1 mark**) Do you expect faster convergence of (13) with $\omega = \frac{1}{2}$ or with $\omega = \frac{2}{3}$?

5. (**2 marks**) Show that, in the particular case where $\omega = \frac{2}{3}$, the iterative scheme (13) coincides with the Newton–Raphson method applied to the nonlinear equation

$$f(x) = 0, \tag{14}$$

for an appropriate function $f\colon \mathbf{R} \to \mathbf{R}$.

6. (**2 marks**) Illustrate graphically a few iterations of the Newton–Raphson method for solving (14) when starting from $x_0 = 2$. You may either create your own figure or write on Figure 1 at the end of this document.

7. *(**Bonus +2**) Prove Theorem 2 in the appendix. More precisely, show that the assumptions of the proposition imply that there is $\delta > 0$ and $L < 1$ such that the following local Lipschitz condition is satisfied:

$$\forall x \in [x_* - \delta, x_* + \delta], \qquad |F(x) - F(x_*)| \leqslant L|x - x_*|. \tag{15}$$

For completeness, one should then show that (15) is sufficient to guarantee local exponential stability, but this is taken for granted here; you do not need to prove this.

Solution. **1.** Since $x_*^3 = 2$, it holds that $\frac{2}{x_*^2} = x_*$, and so $F(x_*) = \omega x_* + (1-\omega)x_* = x_*$.

2.
1: Choose $\omega \in (0,1)$, $x_0 > 0$, a tolerance $\varepsilon > 0$ and a maximum number of iterations K.
2: **for** $k \in \{0, 1, \ldots, K-1\}$ **do**
3: $\quad x_{k+1} \leftarrow \omega x_k + (1-\omega)\frac{2}{x_k^2}$
4: $\quad$ **if** $|x_{k+1} - x_k| < \varepsilon$ **then**
5: $\quad\quad$ **return** x_{k+1}
6: $\quad$ **end if**
7: **end for**

(An alternative stopping criterion is $|x_{k+1}^3 - 2| < \varepsilon$.)

3. For $\omega \in (0,1)$, the function F maps $(0,\infty)$ to $(0,\infty)$ and is continuously differentiable, with $F'(x) = \omega - 4(1-\omega)x^{-3}$. Therefore $F'(x_*) = \omega - 2(1-\omega) = 3\omega - 2$, and $|F'(x_*)| < 1$ if and only if $\omega \in \left(\frac{1}{3}, 1\right)$. By Theorem 2, x_* is locally exponentially stable for these values of ω.

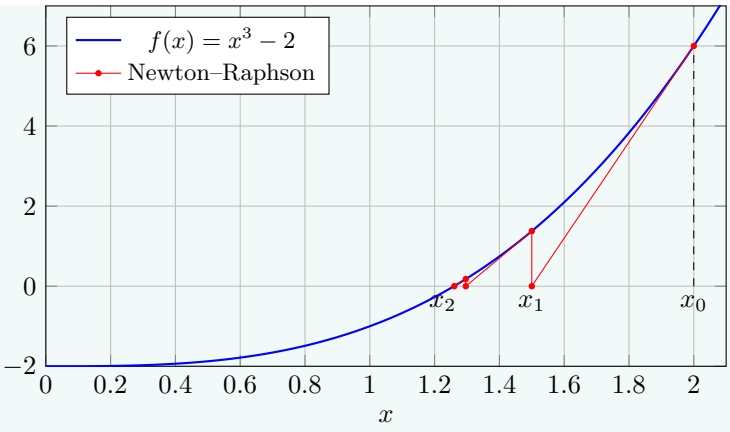
4. For $\omega = \frac{1}{2}$, we have $F'(x_*) = -\frac{1}{2}$, and the convergence is linear: the error is roughly halved at each iteration. For $\omega = \frac{2}{3}$, we have $F'(x_*) = 0$, and the convergence is superlinear (in fact quadratic). We therefore expect faster convergence for $\omega = \frac{2}{3}$.

5. For $f(x) = x^3 - 2$, the Newton–Raphson method reads

$$x_{k+1} = x_k - \frac{x_k^3 - 2}{3x_k^2} = \frac{2}{3}x_k + \frac{1}{3} \times \frac{2}{x_k^2},$$

which is (13) with $\omega = \frac{2}{3}$.

6. x_{k+1} is the intersection with the x-axis of the tangent to the graph of f at $\big(x_k, f(x_k)\big)$. Starting from $x_0 = 2$, we obtain $x_1 = 1.5$, $x_2 \approx 1.2963$, $x_3 \approx 1.2609$, and $x_* \approx 1.2599$:



7. Let $L = \frac{1+|F'(x_*)|}{2} < 1$. Since F' is continuous and $|F'(x_*)| < L$, there exists $\delta \in (0, x_*)$ such that $|F'(x)| \leqslant L$ for all $x \in [x_* - \delta, x_* + \delta]$. For x in this interval, the mean value theorem gives ξ between x and x_* such that

$$|F(x) - F(x_*)| = |F'(\xi)||x - x_*| \leqslant L|x - x_*|.$$

△

Question 6 (Iterative methods for eigenvalue problems, **10 marks**)**.** Let $\|\bullet\|$ denote both the Euclidean norm on vectors and the induced matrix norm. Assume that $\mathsf{A} \in \mathbf{R}^{n\times n}$ is symmetric and nonsingular, and that all the eigenvalues of A have different moduli:

$$|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n|.$$

1. (**5 marks**) Describe with words and pseudocode a simple numerical method for calculating the eigenvalue of A of smallest modulus as well as the corresponding eigenvector.

2. (**2 marks**) Suppose that we have calculated the smallest eigenvalue in modulus λ_n, as well as the associated normalized eigenvector $\boldsymbol{v}_n$. We let

$$\mathsf{B} := \mathsf{A}^{-1} - \frac{1}{\lambda_n}\boldsymbol{v}_n\boldsymbol{v}_n^T.$$

If we apply the power iteration to this matrix, what convergence can we expect? Justify your answer.

3. *(**3 marks**) The aim of this part is to provide an answer to the following question: given an approximate eigenpair $(\widehat{\boldsymbol{v}}, \widehat{\lambda})$, what is the smallest perturbation E that we need to apply to A in order to guarantee that $(\widehat{\boldsymbol{v}}, \widehat{\lambda})$ is an exact eigenpair, i.e. that

$$(\mathsf{A} + \mathsf{E})\widehat{\boldsymbol{v}} = \widehat{\lambda}\widehat{\boldsymbol{v}}\,?$$

Assume that $\widehat{\boldsymbol{v}}$ is normalized and let $\mathcal{E} = \left\{ \mathsf{E} \in \mathbf{C}^{n\times n} : (\mathsf{A}+\mathsf{E})\widehat{\boldsymbol{v}} = \widehat{\lambda}\widehat{\boldsymbol{v}} \right\}$. Prove that

$$\min_{\mathsf{E}\in\mathcal{E}} \|\mathsf{E}\| = \|\boldsymbol{r}\|, \qquad \boldsymbol{r} := \mathsf{A}\widehat{\boldsymbol{v}} - \widehat{\lambda}\widehat{\boldsymbol{v}}. \tag{16}$$

Hint: You may find it useful to proceed as follows:

- Show first that $\mathsf{E} \in \mathcal{E}$ if and only if $\mathsf{E}\widehat{\boldsymbol{v}} = -\boldsymbol{r}$.
- Deduce from the previous item that

$$\forall \mathsf{E} \in \mathcal{E}, \qquad \|\mathsf{E}\| \geqslant \|\boldsymbol{r}\|.$$

- Find a rank one matrix $\mathsf{E}_* \in \mathcal{E}$ such that $\|\mathsf{E}_*\| = \|\boldsymbol{r}\|$, and then conclude. Recall that any rank 1 matrix can be written in the form $\mathsf{E}_* = \boldsymbol{u}\boldsymbol{w}^*$, with norm $\|\boldsymbol{u}\|\|\boldsymbol{w}\|$.

4. (**Bonus +2**) Suppose that we have calculated λ_n and λ_{n-1} together with the associated normalized eigenvectors. Propose a method for calculating the third smallest eigenvalue in modulus, i.e. λ_{n-2}.

Solution. Since A is symmetric, it has an orthonormal basis of eigenvectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$, with $\mathsf{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$.

1. We can use the *inverse iteration*, i.e. the power iteration applied to A^{-1}. The eigenvalues of A^{-1} are λ_i^{-1}, with the same eigenvectors, and the eigenvalue of largest modulus is λ_n^{-1}. In practice, we do not compute A^{-1}: we solve a linear system at each iteration, for example using an LU (or Cholesky, if A is positive definite) decomposition of A computed once and for all.

1: Choose $\boldsymbol{x}_0$ with $\|\boldsymbol{x}_0\| = 1$
2: **for** $k \in \{0, 1, \dots\}$ **do**
3: Solve $\mathsf{A}\boldsymbol{y} = \boldsymbol{x}_k$
4: $\boldsymbol{x}_{k+1} \leftarrow \boldsymbol{y}/\|\boldsymbol{y}\|$
5: $\lambda^{(k+1)} \leftarrow \boldsymbol{x}_{k+1}^T \mathsf{A}\boldsymbol{x}_{k+1}$ ▷ Rayleigh quotient
6: **end for**

If $\boldsymbol{v}_n^T \boldsymbol{x}_0 \neq 0$, the iterate $\boldsymbol{x}_k$ converges (up to sign) to $\pm\boldsymbol{v}_n$ and $\lambda^{(k)}$ converges to λ_n, with an error of order $\left|\lambda_n/\lambda_{n-1}\right|^k$ for the eigenvector. A stopping criterion can be based on the residual $\|\mathsf{A}\boldsymbol{x}_k - \lambda^{(k)}\boldsymbol{x}_k\|$.

2. Since the eigenvectors are orthonormal, $\mathsf{B}\boldsymbol{v}_n = \lambda_n^{-1}\boldsymbol{v}_n - \lambda_n^{-1}\boldsymbol{v}_n = 0$, and $\mathsf{B}\boldsymbol{v}_i = \lambda_i^{-1}\boldsymbol{v}_i$ for $i < n$. The eigenvalue of B of largest modulus is therefore λ_{n-1}^{-1}, and the second largest in modulus is λ_{n-2}^{-1}. Therefore, if $\boldsymbol{v}_{n-1}^T \boldsymbol{x}_0 \neq 0$, the power iteration applied to B converges to $\pm\boldsymbol{v}_{n-1}$ (and the Rayleigh quotient to λ_{n-1}^{-1}), with an error of order $\left|\lambda_{n-1}/\lambda_{n-2}\right|^k$.

3.

- By definition, $\mathsf{E} \in \mathcal{E}$ if and only if $\mathsf{E}\widehat{\boldsymbol{v}} = \widehat{\lambda}\widehat{\boldsymbol{v}} - \mathsf{A}\widehat{\boldsymbol{v}} = -\boldsymbol{r}$.
- If $\mathsf{E} \in \mathcal{E}$, then $\|\boldsymbol{r}\| = \|\mathsf{E}\widehat{\boldsymbol{v}}\| \leqslant \|\mathsf{E}\|\|\widehat{\boldsymbol{v}}\| = \|\mathsf{E}\|$.
- Let $\mathsf{E}_* = -\boldsymbol{r}\widehat{\boldsymbol{v}}^*$. Then $\mathsf{E}_*\widehat{\boldsymbol{v}} = -\boldsymbol{r}(\widehat{\boldsymbol{v}}^*\widehat{\boldsymbol{v}}) = -\boldsymbol{r}$, so $\mathsf{E}_* \in \mathcal{E}$, and $\|\mathsf{E}_*\| = \|\boldsymbol{r}\|\|\widehat{\boldsymbol{v}}\| = \|\boldsymbol{r}\|$.

The minimum in (16) is therefore attained at E_* and equal to $\|\boldsymbol{r}\|$.

4. We can use deflation again, and apply the power iteration to the matrix

$$\mathsf{C} = \mathsf{A}^{-1} - \frac{1}{\lambda_n}\boldsymbol{v}_n\boldsymbol{v}_n^T - \frac{1}{\lambda_{n-1}}\boldsymbol{v}_{n-1}\boldsymbol{v}_{n-1}^T,$$

whose eigenvalues are 0 (twice) and λ_i^{-1} for $i \leqslant n-2$. The eigenvalue of largest modulus is λ_{n-2}^{-1}. Alternatively, we can apply the power iteration to A^{-1} while orthogonalizing the iterates against $\boldsymbol{v}_n$ and $\boldsymbol{v}_{n-1}$ at each step.

△

Auxiliary results

Proposition 1. *Assume that $f\colon [a,b] \to \mathbf{R}$ is a function in $C^2([a,b])$ and let $\widehat{f}$ denote the interpolation of f at two distinct interpolation nodes y_1, y_2. Then there exists $\xi\colon [a,b] \to [a,b]$ such that*

$$\forall y \in [a,b], \qquad f(y) - \widehat{f}(y) = \frac{f''\big(\xi(y)\big)}{2}(y-y_1)(y-y_2).$$

Proposition 2. *Assume that $F\colon (0,\infty) \to (0,\infty)$ is continuously differentiable, and suppose that $x_* \in (0,\infty)$ is a fixed point of the iteration $x_{k+1} = F(x_k)$. If*

$$|F'(x_*)| < 1,$$

then the fixed point x_* is locally exponentially stable.

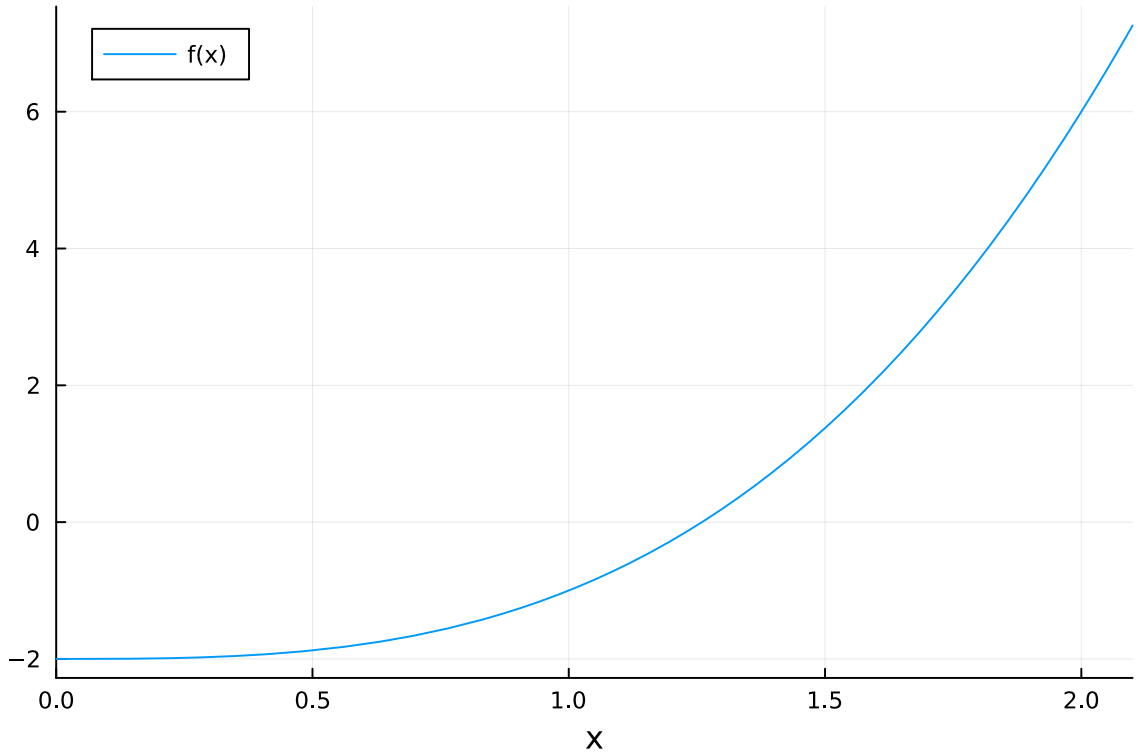


Figure 1: You can use this figure to illustrate the Newton–Raphson method.